the terms *real* and *imaginary* are just names used to keep the two separate. Now that complex numbers have been introduced, it should be quite obvious were the names come from. For example, imagine a 1024 point signal being decomposed into 513 cosine waves and 513 sine waves. Using substitution, we can represent the spectrum by 513 complex numbers. However, don't be misled into thinking that this is the *complex Fourier transform*, the topic of Chapter 31. This is still the *real Fourier transform*; the spectrum has just been placed in a complex format by using substitution.

Electrical Circuit Analysis

This method of substituting complex numbers for cosine & sine waves is called the **Phasor transform**. It is the main tool used to analyze networks composed of resistors, capacitors and inductors. [More formally, electrical engineers define the phasor transform as multiplying by the complex term: $e^{j\omega t}$ and taking the real part. This allows the procedure to be written as an equation, making it easier to deal with in mathematical work. "Substitution" achieves the same end result, but is less elegant].

The first step is to understand the relationship between the current and voltage for each of these devices. For the resistor, this is expressed in Ohm's law: $v = iR$, where i is the instantaneous current through the device, v is the instantaneous voltage across the device, and R is the resistance. In contrast, the capacitor and inductor are governed by the differential equations: $i = C\,dv/dt$, and $v = L\,di/dt$, where C is the capacitance and L is the inductance. In the most general method of circuit analysis, these nasty differential equations are combined as dictated by the circuit configuration, and then solved for the parameters of interest. While this will answer *everything* about the circuit, the math can become a real mess.

This can be greatly simplified by restricting the signals to be sinusoids. By representing these sinusoids with complex numbers, the difficult *differential* equations can be directly replaced with much simpler *algebraic* equations. Figure 30-5 illustrates how this works. We treat each of these three components (resistor, capacitor & inductor) as a *system*. The input to the system is the sinusoidal current through the device, while the output is the sinusoidal voltage across its two terminals. This means we can represent the input and output of the system by the two complex variables: I (for current) and V (for voltage), respectively. The relation between the input and output can also be expressed by a complex number. This complex number is called the **impedance**, and is given the symbol: Z. This means:

$$I \times Z = V$$

In words, the complex number representing the sinusoidal voltage is equal to the complex number representing the sinusoidal current multiplied by the impedance (another complex number). Given any two, the third can be